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Topic 11 — Nuclear Radiation

11.1 Types of Nuclear Radiation

Radioactive decay: The spontaneous and random disintegration of an unstable nucleus, resulting in the emission of radiation and the formation of a different nuclide. "Spontaneous" means it cannot be influenced by external conditions (temperature, pressure, chemical state). "Random" means it is impossible to predict which particular nucleus will decay next, or when.

Properties of Alpha, Beta, and Gamma Radiation

Property	Alpha (α)	Beta-minus (β^-)	Beta-plus (β^+)	Gamma (γ)
Identity	Helium-4 nucleus (^4_2He)	Electron ($^0_{-1}e$)	Positron ($^0_{+1}e$)	EM photon
Charge	+2e	−1e	+1e	0
Mass (u)	4	~1/1836	~1/1836	0
Speed	~5% of c	Up to ~99% of c	Up to ~99% of c	c (speed of light)
Ionising ability	Very high (~10^4 ion pairs/mm)	Moderate (~10^2 ion pairs/mm)	Moderate	Very low (~1 ion pair/mm)
Penetrating power	Low (stopped by paper/skin/few cm air)	Medium (stopped by ~3 mm aluminium)	Medium	High (reduced by thick lead/concrete)

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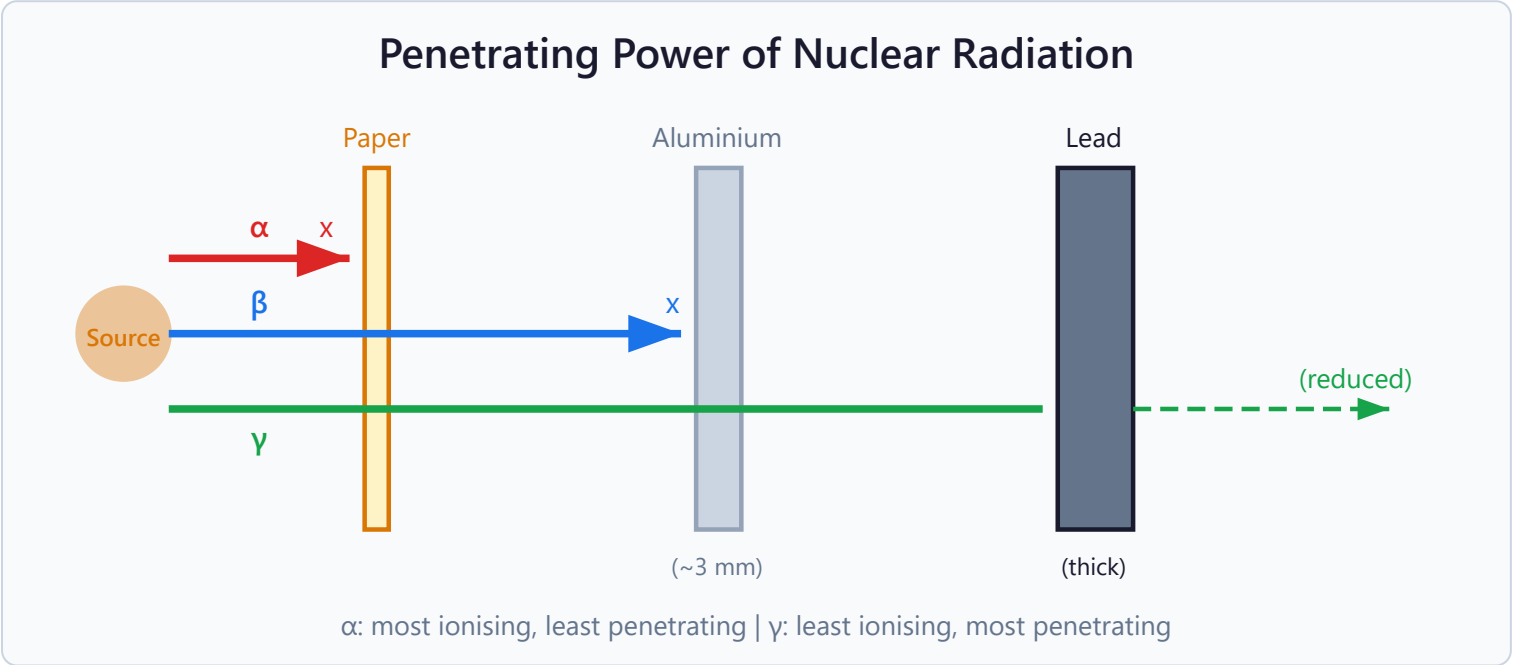
Deflection in E/B fields	Slight (large mass, +2 charge)	Large (opposite direction to α)	Large (same direction as α)	None
Range in air	~5 cm	~1 m	~1 m	Effectively infinite (intensity decreases)



Exam Tip: Ionising ability and penetrating power are inversely related. Alpha particles ionise strongly because of their large charge and mass, so they lose energy quickly and cannot penetrate far. Gamma photons ionise very weakly, so they can pass through large thicknesses of material. This inverse relationship is a frequently tested concept.

Why are alpha particles the most ionising? They have charge +2e and move relatively slowly. Their strong electric field interacts with many orbital electrons as they pass through matter, pulling them away from atoms. Each ionisation event removes kinetic energy from the alpha particle, so it stops after only a few centimetres in air.

11.2 Nuclear Decay Equations

Alpha Decay

The nucleus emits a helium-4 nucleus. The nucleon number decreases by 4 and the proton number decreases by 2.

$$^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

Example: $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$

Beta-minus (β^-) Decay

A neutron in the nucleus converts to a proton, emitting an electron and an electron antineutrino. The nucleon number is unchanged; the proton number increases by 1.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}_e$$

At the quark level: $d \rightarrow u + e^- + \bar{\nu}_e$ (a down quark changes to an up quark via the weak interaction, mediated by a W^- boson).

Example: $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e + \bar{\nu}_e$

Beta-plus (β^+) Decay

A proton in the nucleus converts to a neutron, emitting a positron and an electron neutrino. The nucleon number is unchanged; the proton number decreases by 1.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu_e$$

Gamma Emission

After alpha or beta decay, the daughter nucleus may be in an excited state. It de-excites by emitting a gamma photon. Neither nucleon number nor proton number changes. Gamma emission often accompanies alpha or beta decay.

> **Exam Tip:** Always check that both nucleon number (top) and proton number (bottom) are conserved in decay equations. Do not forget the neutrino or antineutrino in beta decay. The neutrino

was postulated by Pauli in 1930 to explain the continuous energy spectrum of beta particles (if only two products, beta particles would all have the same energy).

11.3 Radioactive Decay — Half-life and Activity

Half-life (t½): The time taken for half of the radioactive nuclei in a sample to decay, or equivalently, the time for the activity of a sample to fall to half its initial value. The half-life is constant for a given isotope and is not affected by temperature, pressure, or chemical bonding.

Activity (A): The rate of decay — the number of disintegrations per unit time. Measured in becquerels (Bq), where 1 Bq = 1 decay per second.

Decay constant (λ): The probability of a given nucleus decaying per unit time. Units: s^{-1}.

Key Equations

$$A = \lambda N$$

Activity equals the decay constant multiplied by the number of undecayed nuclei.

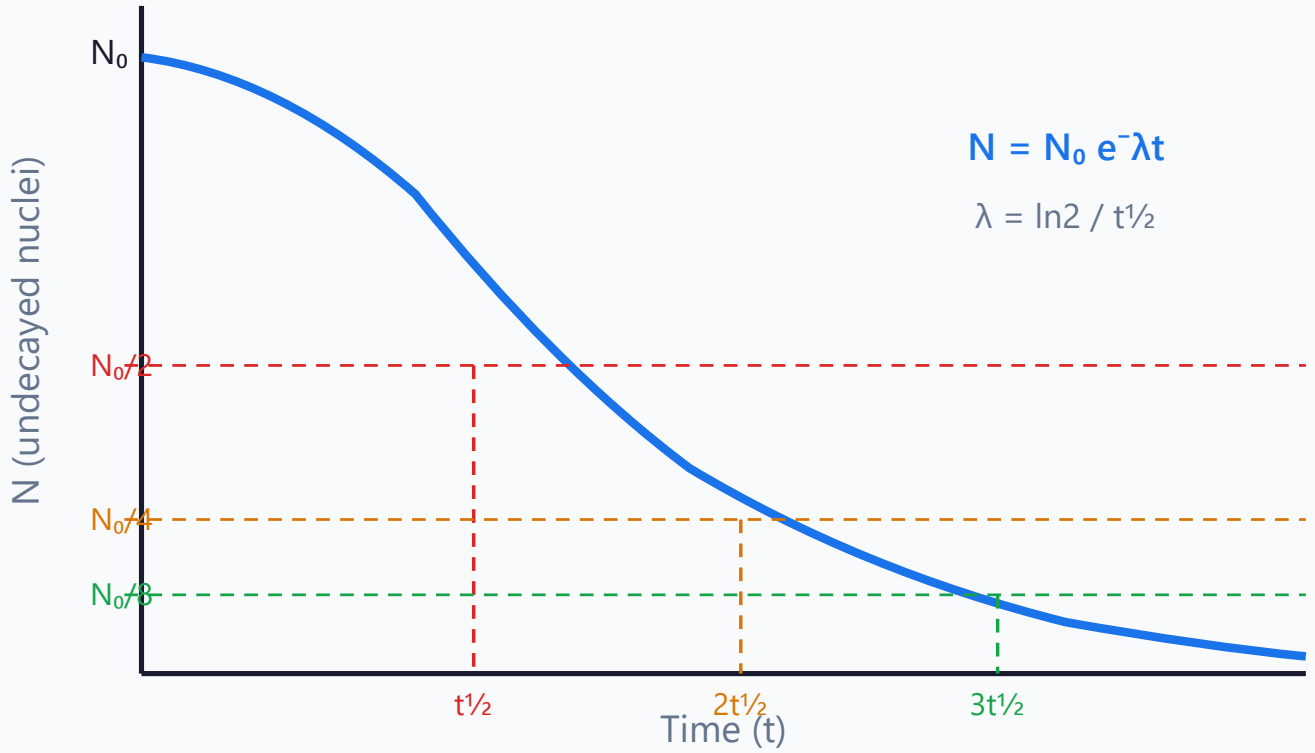
$$\lambda = \ln 2 \,/\, t_{½} \approx 0.693 \,/\, t_{½}$$

$$N = N_0 \, e^{-\lambda t}$$

$$A = A_0 \, e^{-\lambda t}$$

These show that radioactive decay is **exponential**. The number of undecayed nuclei (and the activity) decreases by the same fraction in equal time intervals.

Radioactive Decay: N vs Time



Worked Example: A radioactive source has an initial activity of 8000 Bq and a half-life of 6.0 hours. Calculate: (a) the decay constant, (b) the activity after 15 hours, (c) the time for the activity to fall to 500 Bq.

(a) $\lambda = \ln 2 / t_{½} = 0.693 / (6.0 \times 3600) = 0.693 / 21600 =$ **$3.21 \times 10^{-5}\ s^{-1}$**

(b) $A = A_0 e^{-}\lambda t = 8000 \times e^{-}(3.21 \times 10^{-5} \times 15 \times 3600) = 8000 \times e^{-1.732} = 8000 \times 0.1768 =$ **1410 Bq**

Alternatively: 15 hours = 2.5 half-lives, so $A = 8000 \times (½)^{2.5} = 8000 / 5.66 =$ **1410 Bq**

(c) $500 = 8000\ e^{-}\lambda t$
$e^{-}\lambda t = 500/8000 = 0.0625$
$-\lambda t = \ln(0.0625) = -2.773$
$t = 2.773 / 3.21 \times 10^{-5} = 86\ 400$ s = **24 hours** (= 4 half-lives, which checks: 8000 → 4000 → 2000 → 1000 → 500)

Graphical Determination of Half-life

- A plot of N (or A) against time gives an **exponential decay curve**.
- A plot of ln(N) or ln(A) against time gives a **straight line** with gradient = $-\lambda$.
- After n half-lives, the remaining fraction is $(½)^n$.
- To find t½ from a graph: read the time for the count rate to halve. Take at least three such readings and average for better accuracy.

Exam Tip: If given count rate data, remember to **subtract the background count rate** before analysis. Background radiation comes from cosmic rays, rocks, radon gas, and artificial sources. Measure it with no source present and subtract from all readings.

11.4 Nuclear Fission

Nuclear fission: The splitting of a heavy nucleus (e.g., uranium-235 or plutonium-239) into two smaller daughter nuclei of roughly equal mass, releasing energy, 2–3 neutrons, and gamma radiation.

Fission is induced when a **thermal (slow) neutron** is absorbed by a fissile nucleus. The nucleus becomes highly unstable and splits. The products have higher binding energy per nucleon than the original, so energy is released (typically ~200 MeV per fission event).

Chain Reactions

Each fission event releases 2–3 neutrons. These neutrons can go on to cause further fission events, leading to a **chain reaction**.

- **Uncontrolled chain reaction:** Each fission causes more than one subsequent fission on average — exponential growth of energy release (nuclear weapon).
- **Controlled chain reaction:** Exactly one neutron per fission event causes another fission — steady energy output (nuclear reactor). This is called **criticality**.
- **Sub-critical:** Fewer than one neutron per fission causes another fission — the reaction dies out.

Critical Mass

The minimum mass of fissile material needed to sustain a chain reaction. Below this mass, too many neutrons escape from the surface without causing further fissions.

Nuclear Reactor Components

Component	Function	Examples
Fuel rods	Contain fissile material	Enriched uranium-235 (3–5%), MOX fuel
Moderator	Slows down fast neutrons to thermal speeds (~0.025 eV) so they are more likely to cause fission	Water, heavy water (D_2O), graphite
Control rods	Absorb excess neutrons to maintain exactly one fission per fission event. Raised to increase power, lowered to decrease it.	Boron, cadmium, hafnium
Coolant	Transfers thermal energy from the reactor core to generate steam for turbines	Water, CO_2, liquid sodium
Shielding	Absorbs radiation to protect workers and the environment	Thick concrete, steel, lead

11.5 Nuclear Fusion

Nuclear fusion: The combining of two light nuclei to form a heavier nucleus, releasing energy. This is the process that powers stars. The products have higher binding energy per nucleon, so the mass defect is converted to kinetic energy.

Conditions for Fusion

- **Extremely high temperature** (~10^7–10^8 K) — nuclei need sufficient kinetic energy to overcome the electrostatic (Coulomb) repulsion between positive charges.
- **High density/pressure** — to increase the probability of collisions between nuclei.
- **Sufficient confinement time** — nuclei must be held together long enough for fusion to occur (Lawson criterion).

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These extreme conditions are why controlled fusion for power generation is so difficult. On Earth, experimental reactors such as ITER (under construction in France) and JET (UK) use magnetic confinement to contain plasma at temperatures exceeding 100 million K.

Fusion in Stars

- **Proton–proton (pp) chain:** The dominant fusion process in stars like the Sun ($T \sim 1.5 \times 10^7$ K). Four protons fuse (in a series of steps) to form one helium-4 nucleus, releasing energy, positrons, and neutrinos. Net reaction: $4p \rightarrow He\text{-}4 + 2e^+ + 2\nu_e$ + energy (26.7 MeV).
- **CNO cycle:** Dominates in stars more massive than ~1.3 $M_\odot$ (higher core temperatures). Carbon, nitrogen, and oxygen act as catalysts, also converting four protons to helium-4.

11.6 Binding Energy

Binding energy: The minimum energy required to completely separate all the nucleons in a nucleus into individual free nucleons. Equivalently, the energy released when individual nucleons come together to form a nucleus.

Mass defect (Δm): The difference between the total mass of the individual (separated) nucleons and the actual mass of the nucleus: $\Delta m = Zm_p + Nm_n - M_{nucleus}$. The mass defect is always positive for stable nuclei.

Mass–Energy Equivalence

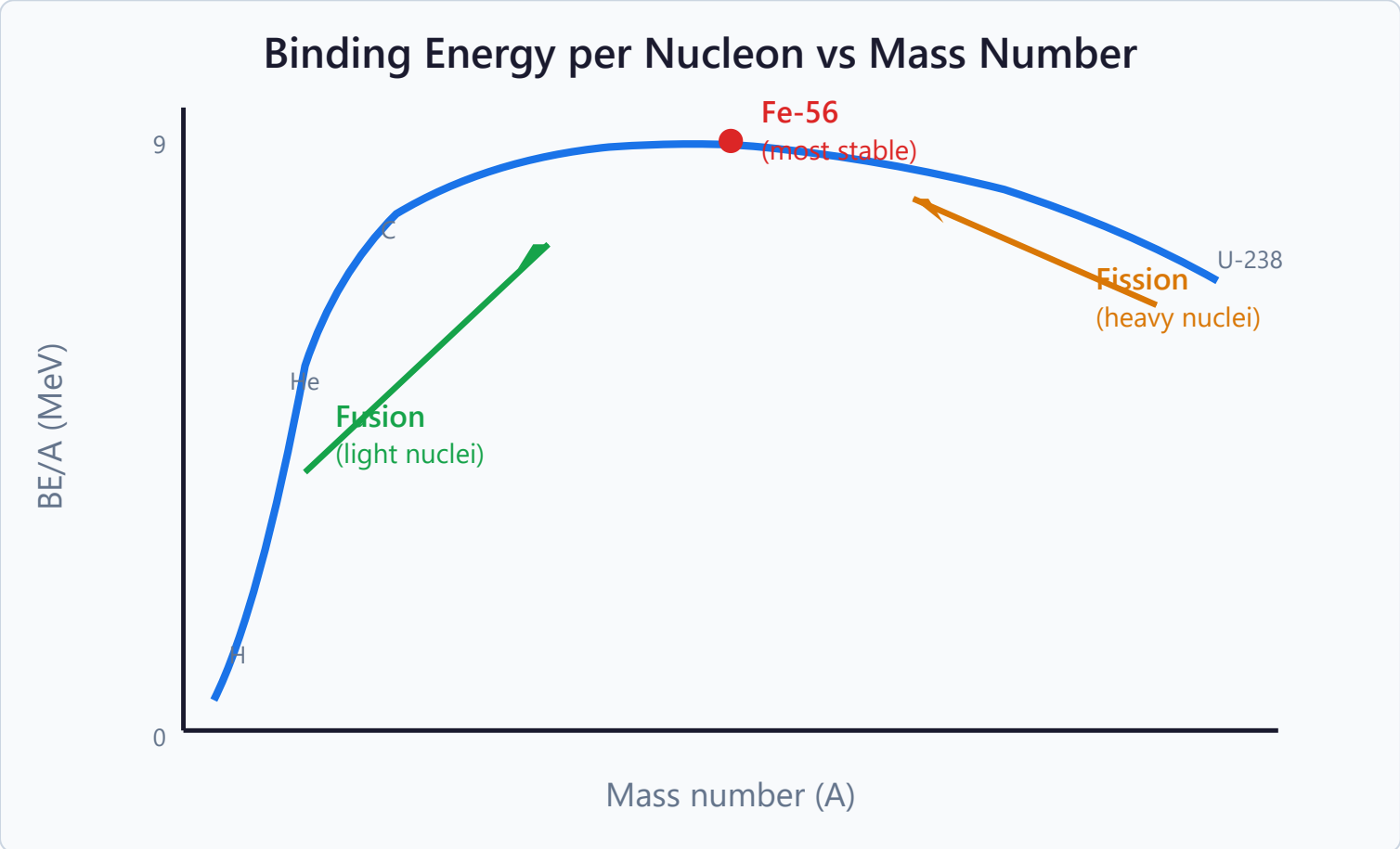
$$E = mc^2$$

The binding energy is related to the mass defect by $E = \Delta m \times c^2$. When nucleons bind together, mass is "lost" and converted to energy. This "missing" mass is the binding energy.

Useful conversion: **1 u = 931.5 MeV/c²**, so a mass defect of 1 u corresponds to a binding energy of 931.5 MeV.

Worked Example: Calculate the binding energy of helium-4, given: m_p = 1.00728 u, m_n = 1.00866 u, M(He-4) = 4.00260 u.

Mass of separate nucleons = 2(1.00728) + 2(1.00866) = 2.01456 + 2.01732 = 4.03188 u
Mass defect = 4.03188 − 4.00260 = 0.02928 u
Binding energy = 0.02928 × 931.5 = **27.3 MeV**
Binding energy per nucleon = 27.3 / 4 = **6.82 MeV per nucleon**



Understanding the BE/A Curve

- The peak occurs at **iron-56** (and nickel-62), which has the highest BE/A (~8.8 MeV per nucleon) and is therefore the most stable nucleus.
- **Fusion** of light nuclei (A < 56) moves up the curve — products have higher BE/A, so energy is released.
- **Fission** of heavy nuclei (A > 56) also moves up the curve — products have higher BE/A, so energy is released.

- Both processes produce products that are **more tightly bound** (more stable), and the difference in binding energy is released.
- Iron-56 cannot release energy by either fusion or fission — it is at the peak. This is why iron accumulates in the cores of massive stars and triggers a supernova.

Calculating Energy Released

For a nuclear reaction, you can calculate energy released in two equivalent ways:

1. **Using masses:** Energy = (mass of reactants − mass of products) × c^2. If mass decreases, energy is released.
2. **Using binding energies:** Energy = total BE of products − total BE of reactants. If BE increases, energy is released.

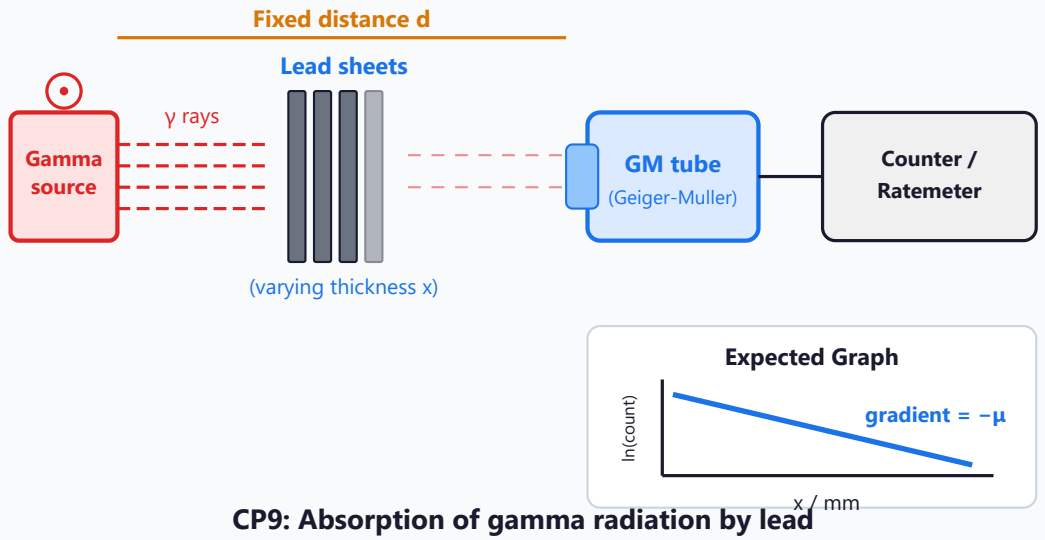
Worked Example: In the fusion reaction $^2H + ^3H \rightarrow ^4He + n$, the masses are: ^2H = 2.01410 u, ^3H = 3.01605 u, ^4He = 4.00260 u, n = 1.00866 u. Calculate the energy released.

Mass of reactants = 2.01410 + 3.01605 = 5.03015 u
Mass of products = 4.00260 + 1.00866 = 5.01126 u
Mass defect = 5.03015 − 5.01126 = 0.01889 u
Energy released = 0.01889 × 931.5 = **17.6 MeV**

This is the energy released per single D-T fusion event — the most promising reaction for future fusion power reactors.

Exam Tip: Energy released = total BE of products − total BE of reactants (when using binding energies). Alternatively, energy released = (mass of reactants − mass of products) × c^2. Be consistent: do not mix the two methods in one calculation. If products are lighter than reactants, the reaction is exothermic.

⚗ Core Practical 9: Investigate the absorption of gamma radiation by lead



CP9: Absorption of gamma radiation by lead

Apparatus

- Sealed gamma source (e.g. cobalt-60 or caesium-137) in a source holder
- Geiger-Muller (GM) tube and counter/ratemeter
- Lead sheets of known, equal thickness (e.g. 1 mm each)
- Ruler (to maintain fixed source-to-detector distance)
- Tongs (for handling the source)
- Stopwatch

Method

1. First, measure the **background count rate**: remove the source to a distant location, count for at least 60 seconds (preferably 300 s), and calculate the background count rate.
2. Set up the gamma source at a **fixed distance** from the GM tube window. Place the lead sheets between source and detector.
3. With **no lead** in place, count for a set time (e.g. 60 s). Record the count and calculate the count rate.
4. Insert 1 lead sheet. Count for the same time. Record.
5. Add more lead sheets one at a time (2, 3, 4, 5, 6... sheets). Record the count rate each time.

6. For each reading, calculate the **corrected count rate** = measured count rate − background count rate.
7. Plot **ln(corrected count rate)** (y-axis) against **thickness x** (x-axis).

Safety

- **Handle the source with tongs at arm's length** — never with bare hands
- Keep the source in its lead-lined box when not in use
- Minimise exposure time — work quickly and efficiently
- Maximise distance from the source when not taking readings
- Do not point the source at anyone
- A trained supervisor should handle the source and maintain a usage log

Analysis

Gamma radiation is absorbed exponentially:

$$I = I_0 e^{-\mu x}$$

Taking natural logs: $\ln(I) = \ln(I_0) - \mu x$

A graph of ln(corrected count rate) vs thickness x gives a straight line with gradient = $-\mu$ (the linear attenuation coefficient) and y-intercept = $\ln(I_0)$.

The **half-value thickness** (HVT) is the thickness that reduces the intensity to half: $x_{½} = \ln 2 / \mu$

Sources of Error & Improvements

Source of Error	Type	Improvement
Statistical fluctuations in count rate (radioactive decay is random)	Random	Count for longer (e.g. 300 s instead of 60 s); repeat and average
Background radiation not subtracted accurately	Systematic	Measure background multiple times and average; count for a long time

Lead sheets not uniform thickness	Systematic	Measure each sheet with a micrometer; use matched sheets
Source-detector distance changes when inserting sheets	Systematic	Fix the source and detector positions; add sheets from the detector side

Exam Tip: The corrected count rate is essential — without subtracting background, the graph will curve at large thicknesses instead of remaining linear on the ln plot. This is because the background becomes a significant fraction of the total count at high absorption.